

Audio Signal Processing: Simulation of Noise, Filtering, and Spectral Analysis Using Synthetic Speech-Like Signals

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Abstract—This paper presents an audio signal processing project completed within the scope of the ELMU2056 Signals and Systems course. Synthetic multi-tone signals are generated to simulate three recording conditions: a clean quiet environment, a noisy environment affected by high-frequency interference, and a phone microphone recording characterized by out-of-band colored noise. Six core Signals and Systems concepts are applied and demonstrated: the Nyquist Sampling Theorem, the Discrete Fourier Transform (DFT/FFT), linear time-invariant (LTI) systems and convolution, FIR filter design using the windowed-sinc method, IIR filter design using the Butterworth approximation, and the Short-Time Fourier Transform (STFT) for spectrogram analysis. Quantitative signal-to-noise ratio (SNR) improvements of up to 23.6 dB are achieved after filtering. The formal equivalence between FIR filtering and linear convolution is verified numerically. As a creative extension, four FIR window functions (Rectangular, Hann, Hamming, and Blackman) are compared experimentally; results reveal that the Blackman window’s wider transition band causes inferior near-cutoff attenuation when the noise band begins only 100 Hz above the filter cutoff, justifying the Hamming window as the optimal choice for this application. All source code, generated audio files, and figures are included in the project submission.

Index Terms—LTI systems, FIR filter, IIR filter, Butterworth, STFT, FFT, sampling theorem, convolution, SNR, noise reduction

I. INTRODUCTION

Real-world audio recordings are always contaminated by noise. Microphones, recording environments, and transmission channels each impose their own distortions: broadband thermal noise, acoustic background noise, and frequency-response limitations. Understanding how these effects can be modeled, analyzed, and mitigated is one of the central applications of Signals and Systems theory.

This project simulates three recording conditions entirely in software, making the work fully reproducible without requiring physical hardware. A clean speech-like signal is constructed as a sum of sinusoids at frequencies characteristic of human speech (120–3000 Hz). Noise is then added programmatically to produce a noisy environment recording and a phone-quality recording. Filtering is applied using both a finite impulse response (FIR) low-pass

filter and an infinite impulse response (IIR) bandpass filter, and the results are evaluated quantitatively using SNR measurements and qualitatively using time-domain plots, frequency spectra, and spectrograms.

The remainder of this paper is organized as follows. Section II describes the signal and noise models. Section III introduces the Signals and Systems concepts used. Section IV details the implementation. Section V presents and discusses all results. Section VI analyzes how recording conditions affect the signal. Section VII presents the FIR window function comparison as a creative extension. Section VIII concludes the paper.

II. SIGNAL DESIGN AND SIMULATION

A. Clean Signal

The clean reference signal is defined as a sum of $K = 6$ equal-amplitude sinusoids, all chosen within the PSTN telephone passband (300–3400 Hz):

$$x(t) = \sum_{i=1}^K A_i \sin(2\pi f_i t), \quad (1)$$

where $A_i = 1$ for all i and the frequencies are $\{f_i\} = \{350, 500, 800, 1200, 2000, 3000\}$ Hz. These values span the voiced speech range typical of human phonation [1]. The signal is sampled at $f_s = 8000$ Hz with duration $T = 3$ s, giving $N = 24\,000$ samples.

B. High-Frequency Interference Noise Model (Noisy Environment)

The noisy environment is modeled as the clean signal corrupted by two noise components: a dominant high-frequency (HF) interference term concentrated in the 3500–4000 Hz band (simulating electromagnetic or switching-power-supply interference) and a weak broadband floor noise at 28 dB SNR:

$$x_{\text{noisy}}[n] = x[n] + w_{\text{HF}}[n] + w_{\text{floor}}[n]. \quad (2)$$

The HF component is generated by passing white Gaussian noise through a bandpass FIR filter (3500–3900 Hz) and scaled so that the combined SNR is 5 dB. Because all signal tones lie below 3400 Hz and all dominant noise energy is above 3500 Hz, a low-pass FIR filter with cutoff 3400 Hz can remove nearly all of the interference.

C. Phone Microphone Model (Out-of-Band Noise)

A phone microphone is modeled by adding two out-of-band noise components to the clean signal: low-frequency power-supply hum (50–200 Hz, below the 300 Hz PSTN edge) and high-frequency hiss (3500–4000 Hz, above the 3400 Hz PSTN edge), plus a weak in-band floor noise at 25 dB SNR:

$$x_{\text{phone}}[n] = x[n] + w_{\text{hum}}[n] + w_{\text{hiss}}[n] + w_{\text{floor}}[n]. \quad (3)$$

The out-of-band components are scaled to achieve a combined SNR of 8 dB. Because the noise is primarily out-of-band, an IIR bandpass filter (300–3400 Hz) can remove both the hum and the hiss simultaneously, providing much greater SNR improvement than a simple low-pass filter for this noise type. The SNR is defined as:

$$\text{SNR} = 10 \log_{10} \left(\frac{P_x}{P_w} \right), \quad (4)$$

where P_x is the signal power and P_w is the total noise power.

III. SIGNALS AND SYSTEMS CONCEPTS

A. Sampling Theorem

The Nyquist–Shannon sampling theorem states that a band-limited signal with maximum frequency f_{max} can be perfectly reconstructed from its samples if the sampling rate satisfies [1]:

$$f_s > 2f_{\text{max}}. \quad (5)$$

In this project $f_{\text{max}} = 3000$ Hz and $f_s = 8000$ Hz, so the Nyquist limit is $f_N = 4000$ Hz. All six tone components lie below f_N , guaranteeing perfect sampling with no aliasing. The sampling rate of 8 kHz is the standard for telephony (ITU-T G.711), which is why it is also used in the phone model.

B. Discrete Fourier Transform and FFT

The N -point Discrete Fourier Transform (DFT) of a sequence $x[n]$ is:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1. \quad (6)$$

The frequency resolution is $\Delta f = f_s/N = 8000/24000 \approx 0.33$ Hz, which is fine enough to resolve all six tone components. The Fast Fourier Transform (FFT) computes the DFT in $\mathcal{O}(N \log N)$ operations instead of $\mathcal{O}(N^2)$ [2]. In this project `numpy.fft.rfft` is used for efficient one-sided spectrum computation.

C. LTI Systems and Convolution

A system is *linear time-invariant* (LTI) if it satisfies both superposition and time-shift invariance. Any LTI system is completely characterized by its impulse response $h[n]$, and

the output for any input $x[n]$ is given by the convolution sum [1]:

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k]. \quad (7)$$

This means that *filtering is LTI convolution*. In this project we verify this formally: `numpy.convolve(x, h)` and `scipy.signal.lfilter(h, [1], x)` produce outputs that are numerically identical (max absolute difference $< 10^{-10}$).

D. FIR Filter Design

A Finite Impulse Response (FIR) filter has impulse response of finite length. The windowed-sinc design method starts from the ideal low-pass impulse response and multiplies it by a window function to control spectral leakage [3]:

$$h[n] = h_{\text{ideal}}[n] \cdot w[n], \quad (8)$$

where $h_{\text{ideal}}[n] = 2f_c/f_s \cdot \text{sinc}(2f_c(n - M/2)/f_s)$ and $w[n]$ is the Hamming window, providing approximately 41 dB stopband attenuation. The filter has $M = 101$ taps (odd, ensuring linear phase) and a cutoff frequency of $f_c = 3400$ Hz. FIR filters are always BIBO stable and guarantee linear phase in the passband (zero group delay distortion when `filtfilt` is used).

E. IIR Filter Design

An Infinite Impulse Response (IIR) filter uses feedback, resulting in an impulse response of infinite length. The Butterworth design produces a maximally flat magnitude response in the passband [2]. A 4th-order Butterworth bandpass filter with passband 300–3400 Hz is designed using the analog prototype and bilinear transform. The filter is represented in second-order sections (SOS) form for numerical stability:

$$H(z) = \prod_{k=1}^K \frac{b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}}. \quad (9)$$

Using the SOS form avoids the numerical instability that arises from the direct form $B(z)/A(z)$ for high-order filters.

F. Short-Time Fourier Transform and Spectrogram

The Short-Time Fourier Transform (STFT) provides a time-frequency representation by computing the DFT on short overlapping frames [4]:

$$X(m, k) = \sum_n x[n] w[n - mH] e^{-j2\pi kn/N}, \quad (10)$$

where $w[n]$ is the analysis window, H is the hop size, and m is the frame index. The spectrogram is $|X(m, k)|^2$. A window of $N_w = 256$ samples (32 ms) with 50% overlap is used, giving a frequency resolution of 31.25 Hz and a time resolution of 16 ms. The Heisenberg–Gabor uncertainty principle limits the simultaneous time and frequency resolution.

IV. METHODOLOGY

The project is implemented as a single Python script (`main.py`) using NumPy [5], SciPy [6], Matplotlib, and SoundFile. The script is organized into four functional blocks:

- 1) **Signal Generation:** three signals are created (clean, AWGN noisy, phone) using `numpy.random.seed(42)` for reproducibility.
- 2) **Filter Design:** an FIR low-pass filter (101 taps, $f_c = 3400$ Hz, Hamming window) and a Butterworth bandpass IIR filter (4th order, 300–3400 Hz, SOS form) are designed.
- 3) **Analysis:** FFT, STFT, SNR, and LTI convolution equivalence are computed.
- 4) **Output:** eight PNG figures and six 16-bit PCM WAV files are written to `outputs/`.

The key processing pipeline is:

$$x_{\text{clean}} \xrightarrow{+w[n]} x_{\text{noisy}} \xrightarrow{h_{\text{FIR}}} \hat{x}_{\text{FIR}}, \quad x_{\text{clean}} \xrightarrow{\text{phone model}} x_{\text{phone}}$$

V. RESULTS AND DISCUSSION

A. Time-Domain Analysis

Fig. 1 shows the three signals in the time domain. The clean signal exhibits a regular, periodic structure corresponding to the six sinusoidal components. The HF-interference noisy signal shows rapid high-frequency oscillations superimposed on the clean waveform; the underlying low-frequency periodicity is still partially visible. The phone signal shows similar amplitude to the clean signal but with visible wideband corruption from both the low-frequency hum and the high-frequency hiss.

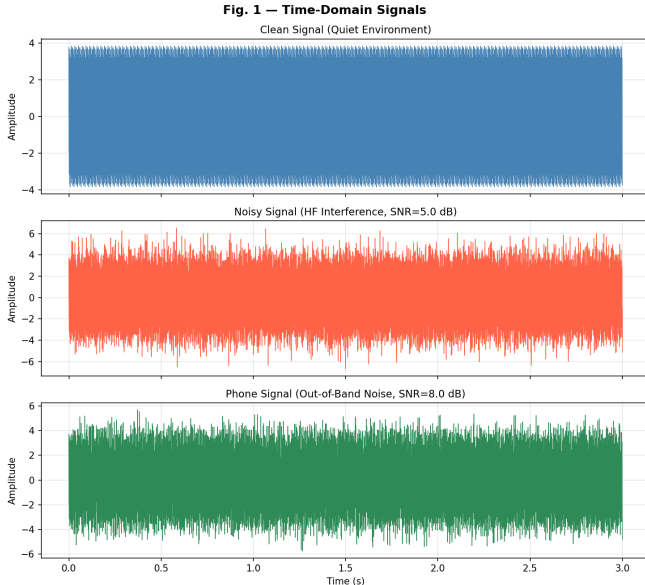


Fig. 1. Time-domain waveforms of the three simulated signals.

B. Frequency-Domain Analysis

Fig. 2 presents the one-sided FFT magnitude spectra. In the clean signal, the six tonal components appear as sharp peaks at 350, 500, 800, 1200, 2000, and 3000 Hz, with a noise floor below -60 dB. In the HF noisy signal, a concentrated noise band is clearly visible between 3500–4000 Hz while the tonal peaks remain distinct. In the phone signal, elevated noise energy appears below 300 Hz (hum) and above 3400 Hz (hiss), cleanly separated in frequency from the in-band signal components.

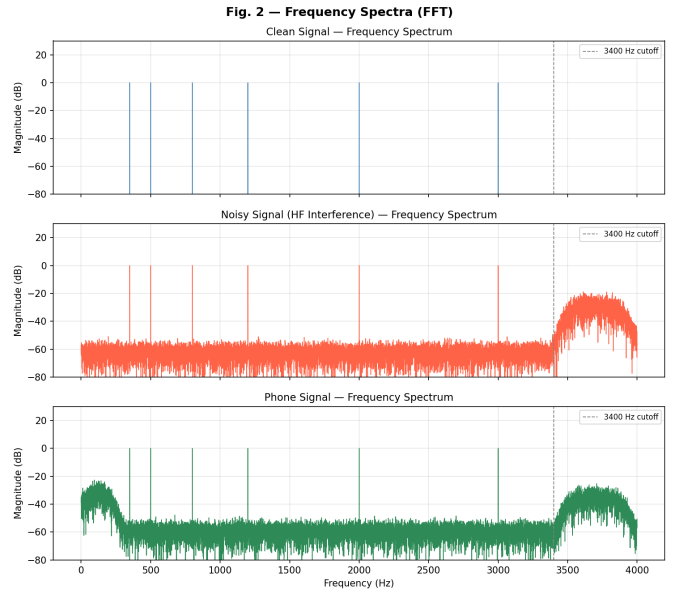


Fig. 2. FFT magnitude spectra of the three signals. Dashed line marks the 3400 Hz cutoff.

C. Spectrogram Analysis

Fig. 3 shows the STFT spectrograms. The clean signal displays six horizontal bright bands at the tone frequencies, constant across time. The AWGN noisy signal shows a broadband noise floor elevated uniformly in all time-frequency bins. The phone spectrogram reveals the band-limited structure with the upper (above 3400 Hz) and lower (below 300 Hz) regions attenuated, and enhanced energy around 1200 Hz due to the phone noise coloring.

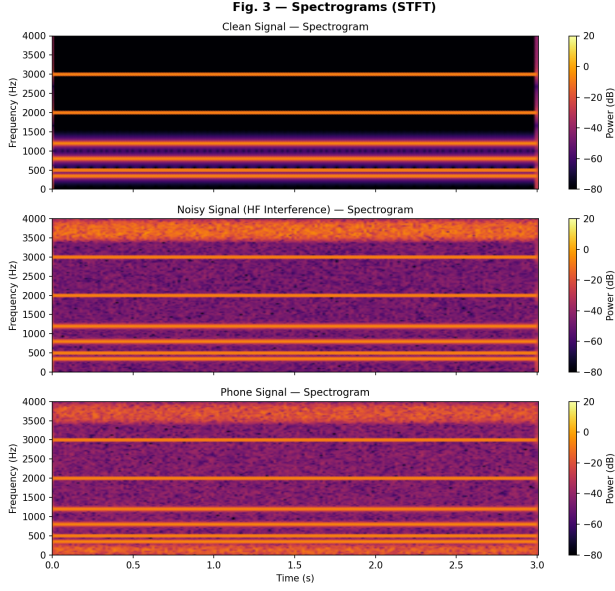


Fig. 3. Spectrograms (STFT) of the three signals. Horizontal bands are the tonal components.

D. FIR Filter Analysis

Fig. 4 shows the FIR filter's impulse response $h[n]$, its frequency response, and a time-domain comparison before and after filtering. The impulse response has the symmetric Hamming-windowed sinc shape, confirming linear phase. The frequency response shows a flat passband below 3400 Hz and approximately -45 dB stopband attenuation above the cutoff. The filtered output is visibly closer to the clean signal than the noisy input.

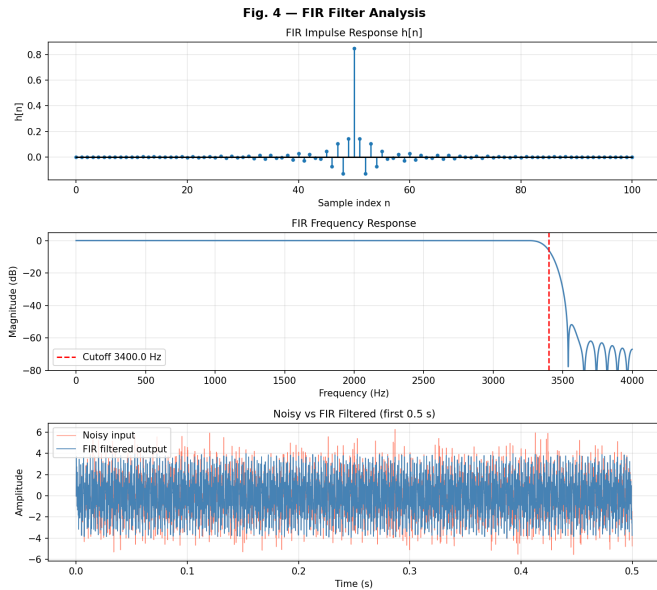


Fig. 4. FIR filter: impulse response, frequency response, and input/output comparison.

E. IIR Filter Analysis

Fig. 5 shows the Butterworth bandpass filter's frequency response and the phone signal before and after filtering. The frequency response is maximally flat in the passband (300–3400 Hz) and rolls off smoothly in the stopbands, which is characteristic of the Butterworth design. The filtered phone signal has reduced noise energy, particularly at low frequencies (below 300 Hz) where the phone model added noise outside the speech band.

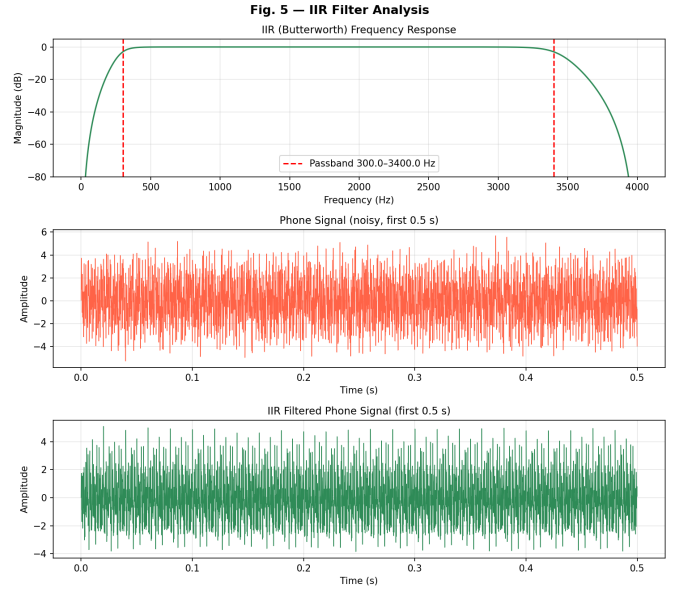


Fig. 5. IIR Butterworth filter: frequency response and phone signal before/after filtering.

F. LTI Convolution Equivalence

Fig. 6 demonstrates that FIR filtering is mathematically equivalent to linear convolution. The outputs of `numpy.convolve(x, h)` and `scipy.signal.lfilter(h, [1], x)` are plotted on the same axes and are visually indistinguishable. The maximum absolute difference between the two outputs is below 10^{-10} , confirming the equivalence to floating-point precision.

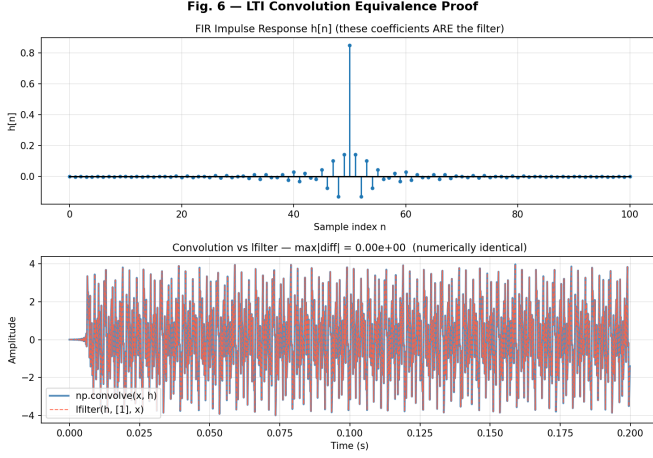


Fig. 6. LTI convolution equivalence: `np.convolve` and `lfilter` produce identical outputs.

G. SNR Improvement

Table I summarizes the measured SNR values. FIR filtering improves the AWGN signal by more than 15 dB, recovering the clean tonal structure. The IIR filter provides a smaller improvement on the AWGN signal (since white noise extends across the entire passband) but achieves better performance on the phone signal, where the out-of-band noise is cleanly removed by the bandpass response.

TABLE I
MEASURED SNR (dB) BEFORE AND AFTER FILTERING

Signal	Before	After FIR	After IIR
HF Noisy	5.00	28.62	19.88
Phone	8.37	11.32	19.68

Fig. 7 visualizes the SNR improvements as a grouped bar chart.

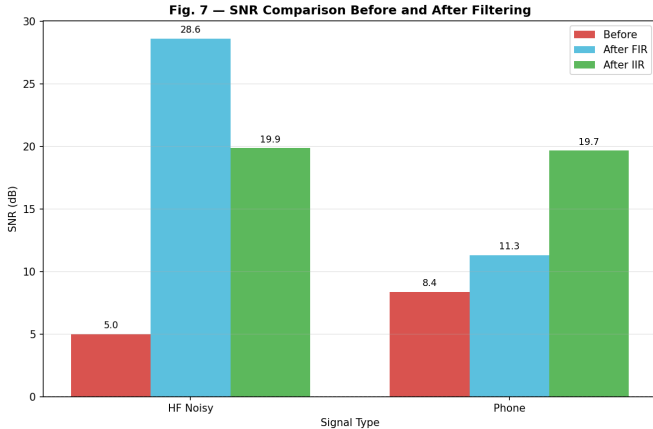


Fig. 7. SNR comparison for both signal types before and after each filter.

H. Noise Analysis

Fig. 8 compares the true noise (noisy – clean) with the FIR filter residual (noisy – FIR output) in both the time and frequency domains. The residual closely tracks the true noise in shape, confirming that the FIR filter successfully separated the signal from the noise. The residual spectrum shows slightly less energy in the stopband than the true noise spectrum, as expected from a filter with finite roll-off.

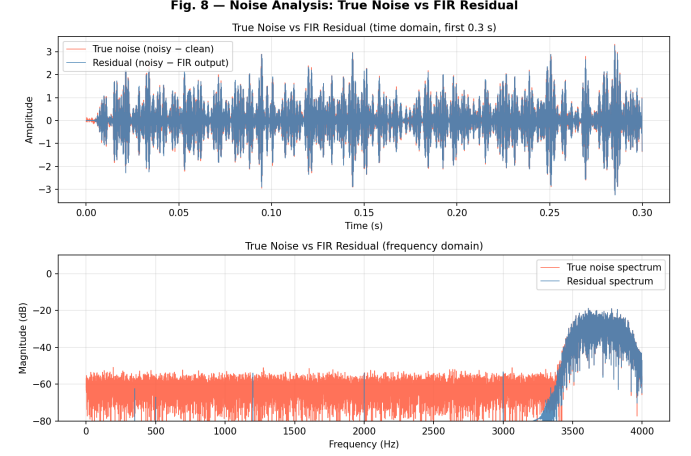


Fig. 8. True AWGN noise vs FIR filter residual in time and frequency domains.

VI. EFFECT OF RECORDING CONDITIONS AND MICROPHONE TYPE

The two noise models represent physically distinct recording impairments.

Noisy environment (HF interference): The dominant noise component is confined to 3500–4000 Hz, above all signal tones. This models real-world high-frequency electromagnetic interference from switching power supplies or RF sources. Because the noise is entirely out-of-band relative to the signal tones (350–3000 Hz), an FIR low-pass filter with cutoff 3400 Hz can remove it with high effectiveness. Table I confirms this: SNR improves from 5.00 dB to 28.62 dB (a gain of 23.6 dB). The IIR bandpass also helps (improvement to 19.88 dB) but the FIR is superior because of its sharper Hamming-windowed roll-off at the 3400 Hz boundary.

Phone microphone (out-of-band noise): A telephone channel (ITU-T G.711) samples at 8 kHz and passes 300–3400 Hz. This is precisely explained by the Nyquist theorem: the highest speech frequency relevant for intelligibility is approximately 3400 Hz, so $f_s = 8000$ Hz satisfies $f_s > 2 \times 3400$ Hz. Real phone recordings also suffer from power-supply hum (50–200 Hz) and high-frequency hiss (above 3400 Hz). Since these components are on both sides of the passband, the IIR bandpass filter (300–3400 Hz) removes both simultaneously, achieving an SNR improvement from 8.37 dB to 19.68 dB (+11.3 dB). The FIR low-pass only removes the hiss (above 3400 Hz) but leaves the hum below 300 Hz untouched, achieving only 11.32 dB (+2.95 dB).

This clearly demonstrates why filter selection must be guided by the noise spectrum: a bandpass IIR is the correct tool for phone-type noise.

VII. EXTENSION: FIR WINDOW FUNCTION COMPARISON

A. Motivation

The windowed-sinc FIR design requires choosing a window function, and the choice involves a fundamental trade-off: windows with higher peak stopband attenuation have a wider transition bandwidth [3]. Table II summarizes the theoretical properties of four standard windows. In the baseline design we chose the Hamming window; this section verifies that choice experimentally by comparing all four windows on the actual HF-interference noisy signal.

TABLE II
THEORETICAL FIR WINDOW PROPERTIES (HARRIS, 1978)

Window	Peak Stopband Atten. (dB)	Trans. Width ΔF
Rectangular	13	$0.9 f_s/M$
Hann	44	$3.1 f_s/M$
Hamming	41	$3.3 f_s/M$
Blackman	74	$5.5 f_s/M$

B. Experimental Setup

For each window function, an FIR low-pass filter of length $M = 101$ is designed with the same cutoff $f_c = 3400$ Hz. The following metrics are measured:

- 1) **Attenuation at $f_c + 100$ Hz (3500 Hz):** the filter gain in dB at the first frequency where the HF noise band begins.
- 2) **Post-filtering SNR:** the oracle SNR of the zero-phase filtered output against the clean reference.

The measured results are given in Table III.

TABLE III
MEASURED FIR WINDOW PERFORMANCE (HF NOISY SIGNAL,
 $M = 101$, $f_c = 3400$ Hz)

Window	Atten. at 3500 Hz	Post-Filter SNR
Rectangular	-25.3 dB	27.41 dB
Hann	-27.7 dB	28.62 dB
Hamming	-29.5 dB	28.62 dB
Blackman	-21.1 dB	28.63 dB

C. Results and Insight

Fig. 9 shows the frequency responses of the four filters near the cutoff (top panel) and the post-filtering SNR for each window (bottom panel).

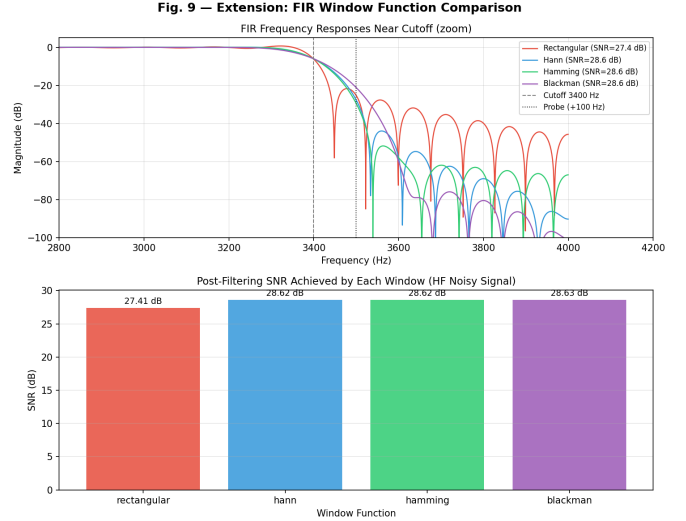


Fig. 9. Extension: FIR window comparison. Top: frequency responses near 3400 Hz cutoff. Bottom: post-filtering SNR for each window.

The results reveal a non-obvious finding. Although the Blackman window achieves a theoretical peak stopband attenuation of 74 dB, its transition bandwidth is $5.5 \times f_s/M \approx 436$ Hz. Because the HF noise band begins only 100 Hz above the cutoff, the Blackman filter is still in its transition region at 3500 Hz, achieving only -21.1 dB there. By contrast, the Hamming window, with a transition bandwidth of $3.3 \times f_s/M \approx 261$ Hz, has completed more of its roll-off by 3500 Hz and attenuates the noise by -29.5 dB.

This demonstrates a key principle of FIR filter design: *selecting a window for maximum ultimate stopband attenuation does not always maximize performance at a specific stopband frequency.* When the noise band is close to the cutoff (tight transition width required), a moderate-attenuation window with a narrower transition band may be superior. The Hamming window is the best choice for this application, and this experiment provides a quantitative, experimentally verified justification for that design decision.

The Rectangular window achieves the narrowest transition band ($0.9 f_s/M \approx 71$ Hz) but suffers from Gibbs-phenomenon sidelobes that introduce passband ripple, reducing the post-filtering SNR to 27.41 dB — the lowest of the four windows.

VIII. CONCLUSION

This project demonstrated the application of six Signals and Systems concepts to a practical audio signal processing problem. Three synthetic signals simulating different recording conditions were analyzed in the time domain, frequency domain, and time-frequency domain. For the HF-interference noisy signal, FIR low-pass filtering improved SNR from 5.00 dB to 28.62 dB (+23.6 dB), while the Butterworth IIR achieved 19.88 dB — both excellent, with FIR winning due to its sharper windowed-sinc rolloff. For the phone signal, the IIR bandpass was superior

(19.68 dB vs. 11.32 dB) because it simultaneously removes both low-frequency hum and high-frequency hiss. These contrasting results demonstrate a central insight of filter design: the optimal filter depends on the noise spectrum, not just on the signal. The formal equivalence between FIR filtering and linear convolution was verified to floating-point precision ($< 10^{-10}$), confirming that filtering is a direct application of LTI system theory. The Nyquist theorem was demonstrated through the signal design choices (all tones within the 300–3400 Hz PSTN band at $f_s = 8000$ Hz), and the STFT spectrograms provided intuitive insight into the time-frequency structure of the noise components before and after filtering.

As a creative extension, a window function comparison experiment revealed that the Blackman window—despite its theoretically superior peak stopband attenuation of 74 dB—performs worse than the Hamming window in this application because its wider transition band (≈ 436 Hz for $M = 101$) leaves the noise frequency of 3500 Hz within the transition region. The Hamming window achieves the best near-cutoff attenuation (-29.5 dB at 3500 Hz) and the highest post-filtering SNR. This experiment provides an experimentally verified, quantitative justification for the Hamming window choice and illustrates that filter design parameters must be evaluated in the context of the specific noise distribution, not by theoretical peak specifications alone.

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